MSc Business Analytics

MGT7222: Human Resources Analytics

Assignment – 2

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1. INTRODUCTION

Losing employees is a tricky problem that so many organizations face these days, no matter what industry they are in. And it's not just about the money it costs to replace people who quit. It also drags down how well the organization performs and how positive the culture feels. When employees leave it's more than just having fewer people on staff - companies have to strategically manage a whole set of challenges that come with turnover.

In a study by Artelt & Gregoriades (2023), it revealed the hidden costs of turnover are high. Beyond recruitment and training expenses, losing staff impacts quality too. At one major Chinese manufacturer, just a 1 percentage point increase in weekly quit rates resulted in a 0.74% to 0.79% rise in defective products. So even in jobs needing few specialized skills, retention pays off by avoiding mistakes. Keeping employees longer, even in roles requiring minimal training, enhances product reliability in ways that are sometimes overlooked.

On top of the money lost, turnover takes a toll in other ways too. Porter (2014) research shows between hiring, training and ramp-up time, replacing an entry-level employee can cost up to what that person makes in a whole year. And that's just playing catch-up—high turnover really drags down customer service and quality since new folks aren't as productive. Over time that can chip away at a company's competitive edge in the market.

Here's the complicated part though - some turnover is actually a good thing. When low performers leave on their own accord, it clears the way for stronger talent to move up and brings opportunities to improve.

So turnover itself isn't necessarily bad or good, which puts leaders in a tricky spot. They've got to balance the costs of turnover and its impact on company culture with the potential gains that can come with change.

That's why we're diving deep into predictive analytics methods that can identify why employees quit. Understanding those drivers better lets companies tailor their retention programs to hold onto top talent. Doing turnover right strengthens performance, culture, and stability over the long-haul.

2. BACKGROUND AND LITERATURE REVIEW

Using data analytics to predict key things about employees represents an important shift in human resources (HR). Whereas HR used to focus more on operations and transactions, it's becoming a real strategic driver for organizations seeking a competitive edge (Opatha, 2020) Predictive analytics refers to various statistical techniques used to forecast future trends. On the simpler end this could mean describing employee satisfaction over time. More advanced methods even identify the variables that drive outcomes like productivity and attrition. Applying predictive analytics allows HR teams to get ahead of challenges instead of just reacting (Gupta, 2020).

For this project, we chose two leading platforms - KNIME and Tableau. KNIME integrates all types of data and runs complex analytics models to uncover insights. Tableau then enables us to translate the findings into impactful, easy-to-understand visuals. Together these tools align perfectly with our goal of discovering what drives attrition at our organization. Identifying those predictors lets leadership shape stronger retention strategies.

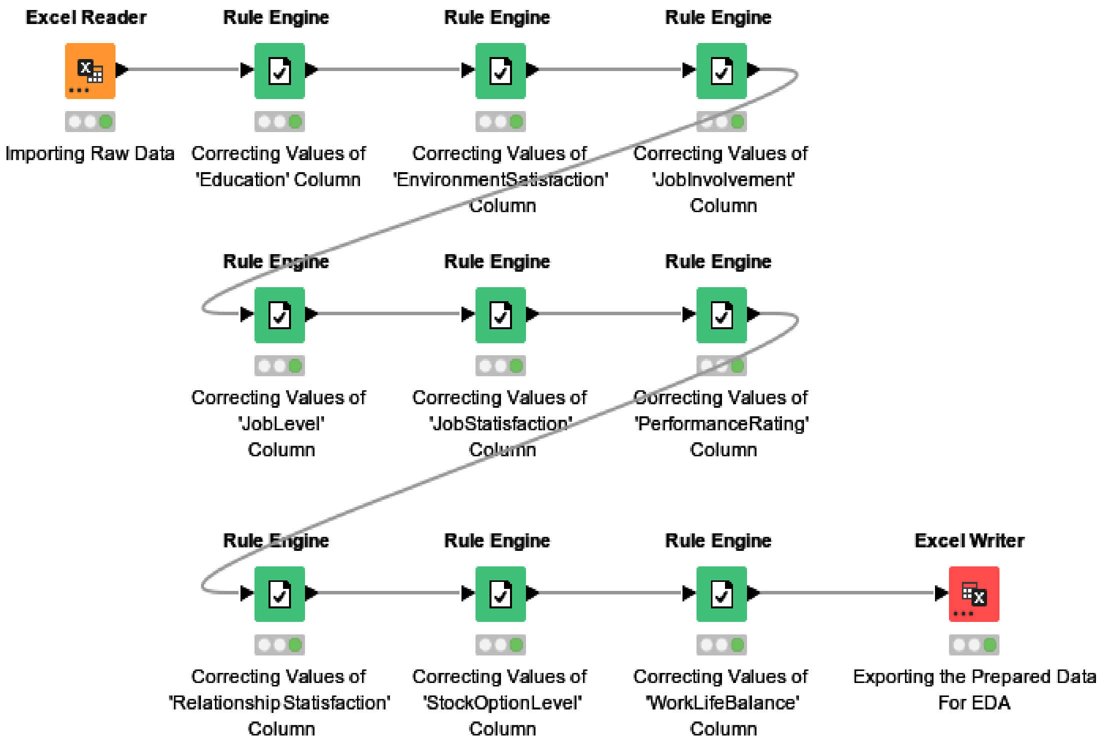
3. METHODOLOGY

The objective of this research is to explore the best model to predict the employee attrition using the various factors responsible for attrition. In order to do so researchers intend to have developed several models. The present segment describes the research methodology of this study. raw data here consists of 35 items related to 1467 employees of the organization. The miscellaneous variables in the dataset are broadly employee related i.e. ID, Age, Gender, etc. and job related like department, JobInvolvement, JobLevel, Satisfaction and most importantly Attrition etc.

The research methodology of this study is broken down in four sections:

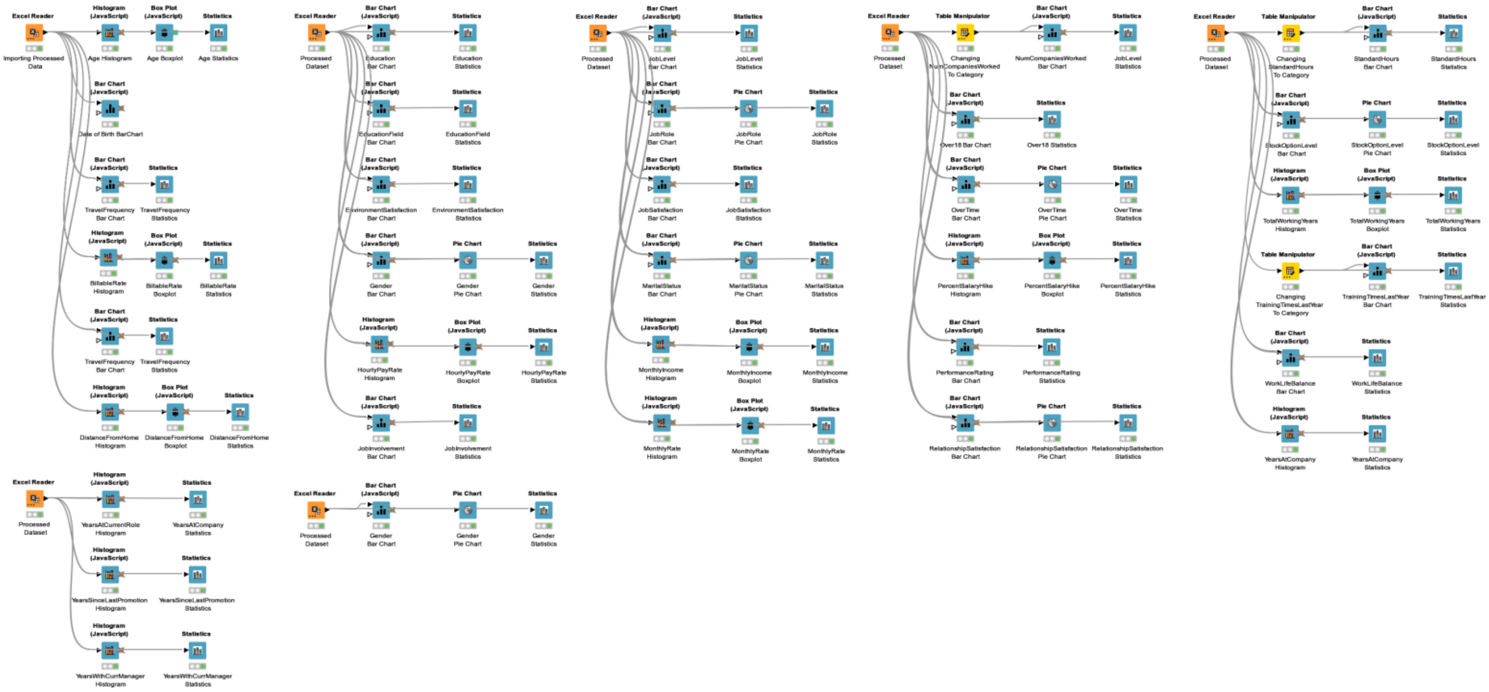
3.1 Data Preparation

This is the first task of any research study to refine the raw data for analysis. To prepare the data an initial inspection of data was conducted, and various fields like Education, JobInvolvement, EnvironmentSatisfaction etc. were corrected in the form of categorical output.

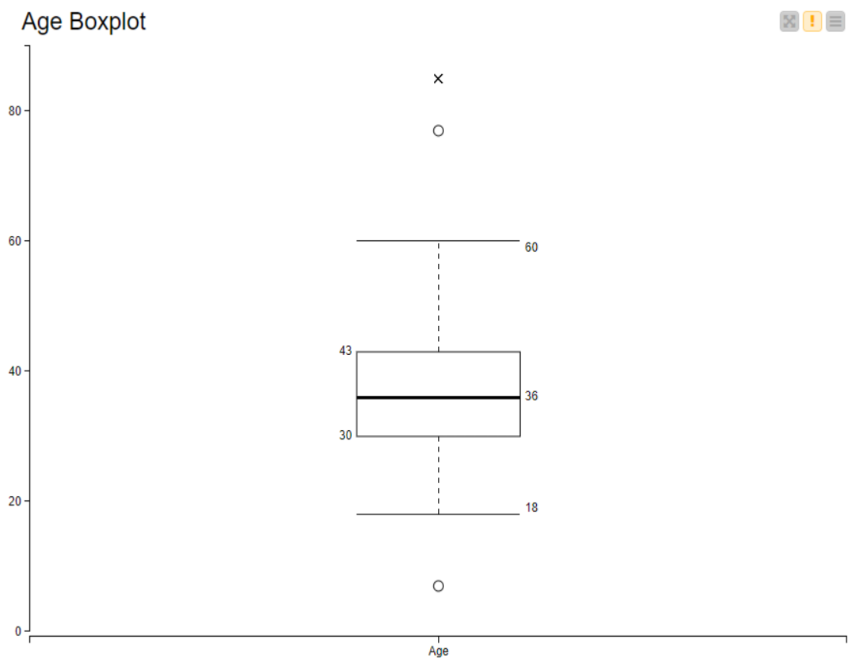


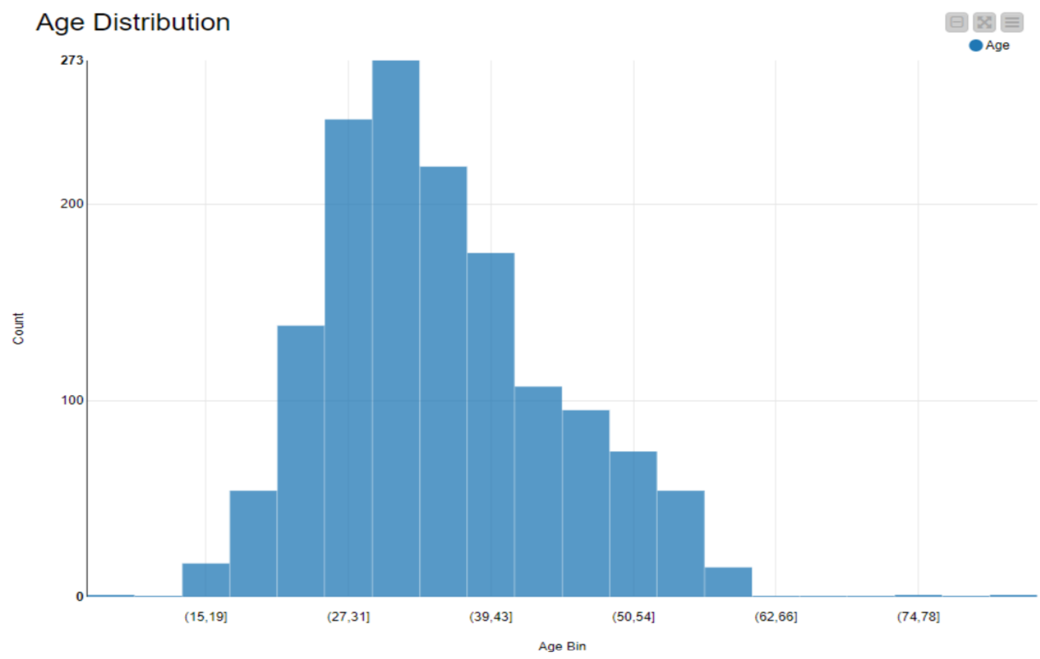
3.2 Data Exploration

To assess the outliers, missing values, and normality, authors used various charts like histogram, boxplot, and summary for Age and YearsAtCurrentRole, and Bar chart for Gender and Education. The KNIME Workflow for the data exploration for various variables is shown below.



Also, the sample output of the plots is following:



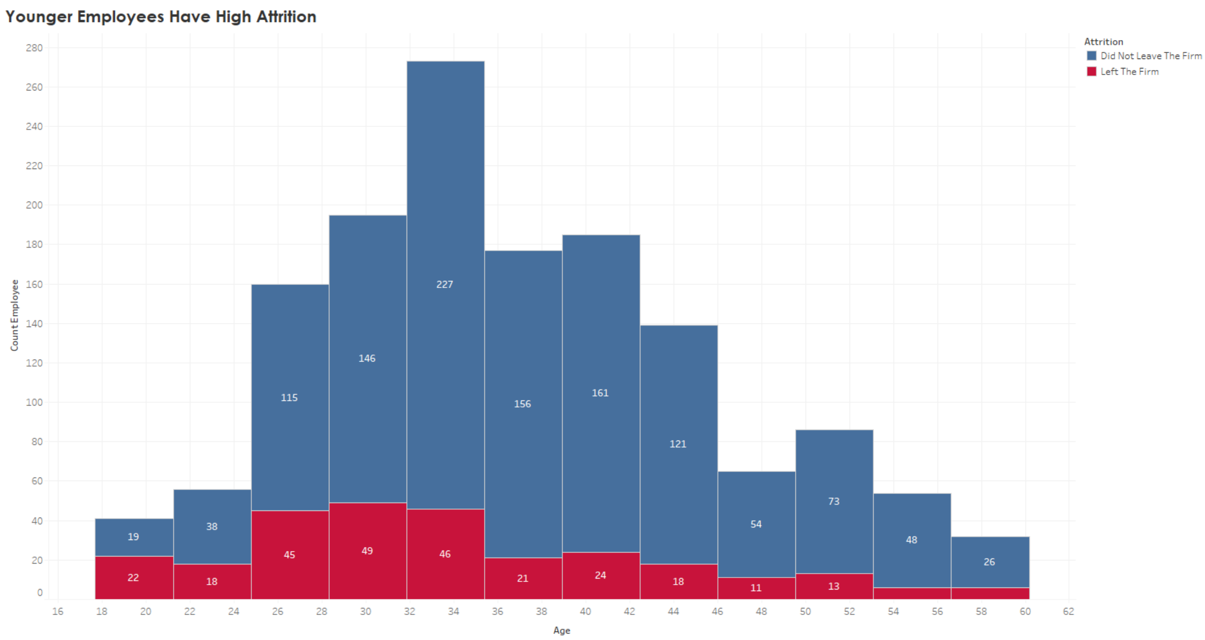


Descriptive Statistics of Age

Rows: 1 | Columns: 15

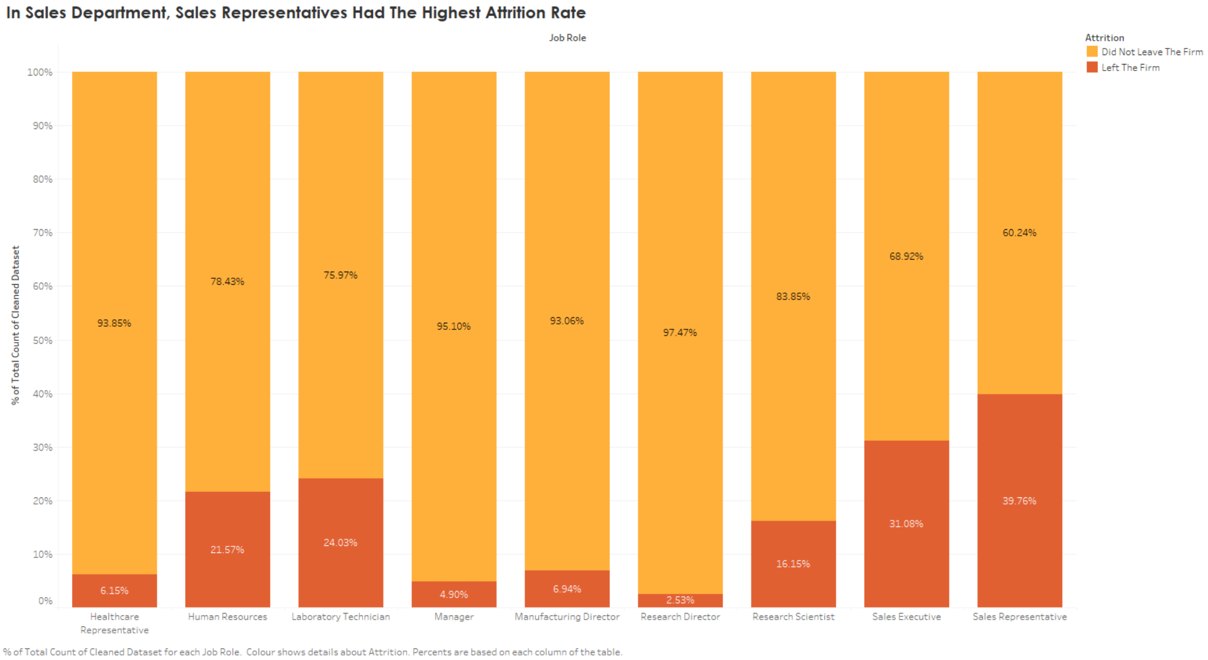
Type	# Missing v...	# Unique val...	Minimum	Maximum	25% Quantile	50% Quantil...	75% Quantile	Mean	Standard De...	Sum	Variance	10 most co...	Skewness
Number (inte...	0	46	7	85	30	36	43	36.967	9.305	54,230	86.575	35 (78; 5.32%...	0.502

Also, authors attempted to explore the attrition against various factors namely the Age, Department, Marital status, and JobRole etc. using Tableau.



The trend of count of Age for Age (bin). Colour shows details about Attrition.

The above chart shows that young employees between 26 to 34 years of age show high attrition.



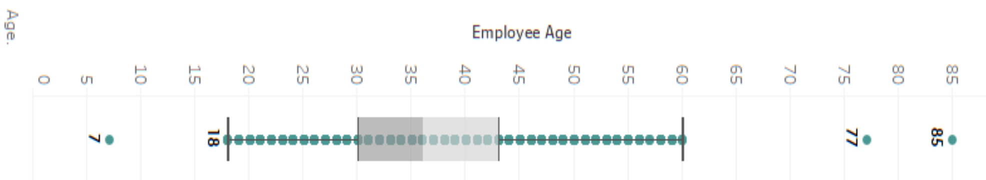
Similarly, the above chart shows sales representatives have highest attrition.

3.3 Data Cleaning

In the third stage, the step of data cleaning takes place after data exploration. To clean the data following steps were carried out:

1) Initially, EnvironmenSatisfaction, JobInvolvement, JobSatisfaction, RelationSatisfaction and WorkLifebalance were aggregated and later a new field 'TotalSatisfaction' was formed by taking mean of this aggregate. The 'TotalSatisfaction' was merged to the original dataset and each of its constituents were removed from the dataset.

2) Three outliers from 'Age' category were removed.



Finding Outliers of 'Age' Using BoxPlot

3) In the next step, the Age and MonthlyIncome were grouped into 'bin' to simplify the density.

4) In the 'TravelFreq' column, the field 'None' was changed as 'Not Travelled'.

5) Few mismatches under 'Department' column were corrected and merged properly.

6) Two columns ‘Over18’ and ‘StandardHours’ were removed because all the entries for the respective columns were same.

7) The datatype for both the columns ‘NumCompaniesWorked’ & ‘TrainingTimeLastYear’ was changed as category.

8) The whole row for total working years equal to 94 was removed.

9) Lastly, the datatypes of several columns were corrected.

10) The above corrected data frame was exported to Tableau and Microsoft Excel for further data analysis.

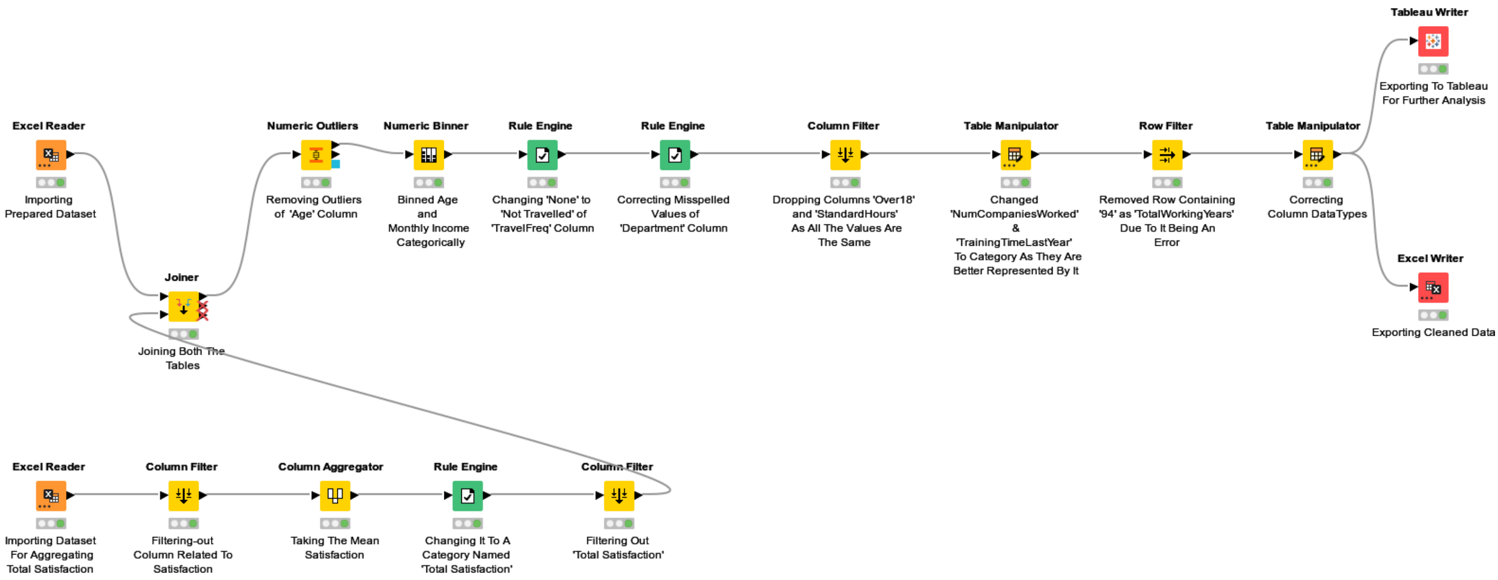


Fig: The KNIME workflow for the data cleaning

3.4 Model Building

In order to prepare a model based on predicting employee attrition, the factors of the model were carefully selected based on correlation analysis. So, the dataset was split into 4 categories as per respective categories namely,

Continuous, Discrete, Ordinal and Nominal.

Since KNIME produces the correlation matrix in the form of a Heatmap and a composite table, whereas the Tableau produces an output in the form of table with correlation statistic and respective p-value. The output from Tableau seems relatively easier to understand. So, the correlation analysis here was carried out based on datatypes of variables using both the KNIME and Tableau.

3.4.1 Correlation between Continuous variables and Attrition

First of all, the correlation between continuous variables and attrition was carried out. To do so, the variable ‘Attrition’ was encoded in the form of 0 (No) and 1(Yes). Following images are the output from KNIME in the form of the HeatMap and composite table for continuous variables.

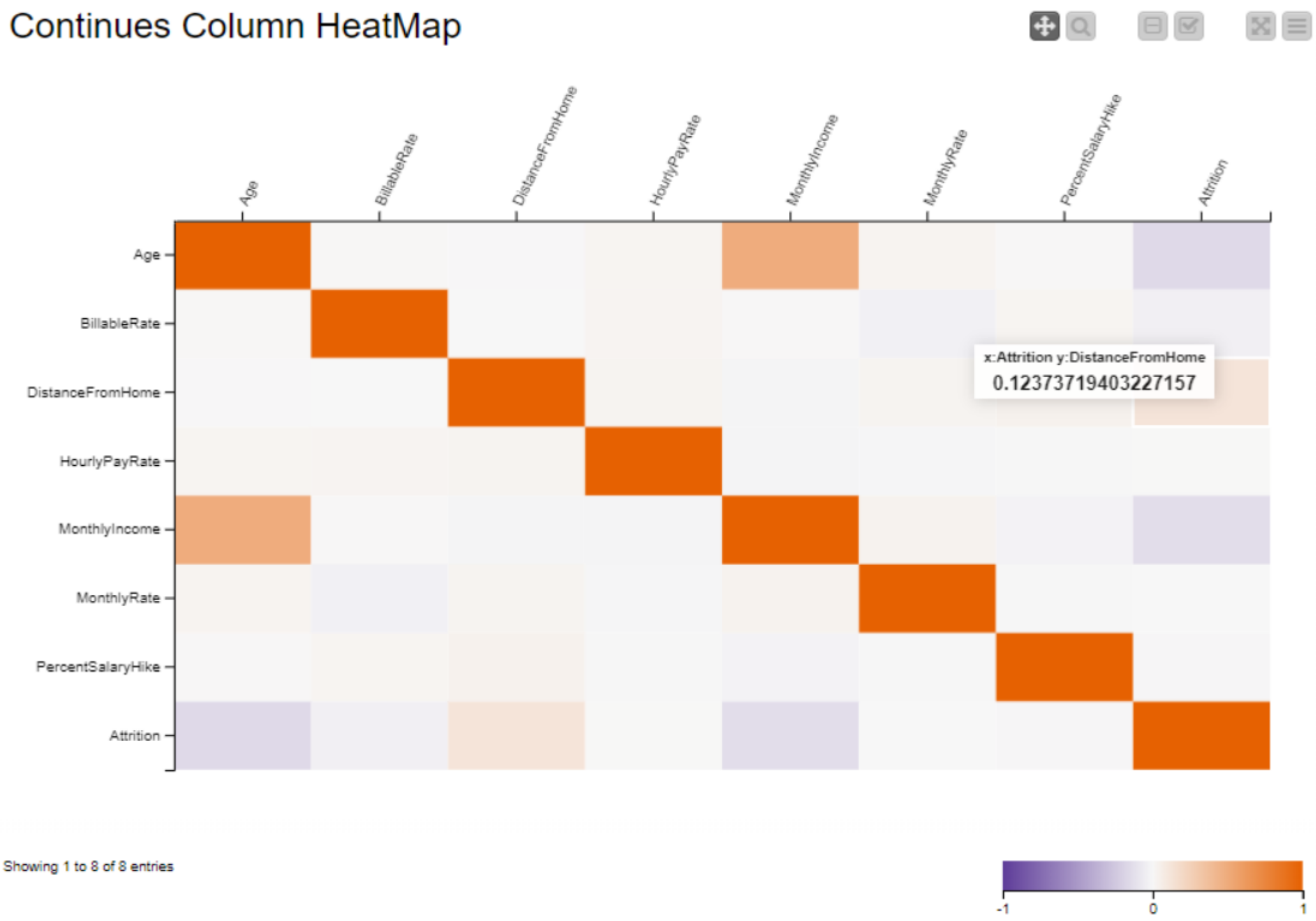


Table "default" - Rows: 28	Spec - Columns: 5	Properties	Flow Variables		
Row ID	S First col...	S Sec...	D Correlation value	D p value	I Degree...
Row6	Age	Attrition	-0.15866127230001523	1.046393066617...	1461
Row12	BillableRate	Attrition	-0.04157950086608568	0.111900177858...	1461
Row17	DistanceFro...	Attrition	0.12373719403227157	2.064003821189...	1461
Row21	HourlyPayRate	Attrition	0.002604479805043...	0.920714099734...	1461
Row24	MonthlyInco...	Attrition	-0.1396029215121 9364	8.257611761033...	1461
Row26	MonthlyRate	Attrition	0.001698510849057...	0.948244727614...	1461
Row27	PercentSala...	Attrition	-0.009337091327134...	0.721212612990...	1461
Row0	Age	BillableRate	0.009895553981058...	0.705294953627...	1461
Row1	Age	DistanceFro...	-0.002878817738144...	0.912394674214...	1461
Row7	BillableRate	DistanceFro...	-0.002336855078363...	0.928838312580...	1461
Row2	Age	HourlyPayRate	0.02249541178623444	0.389896834895...	1461
Row8	BillableRate	HourlyPayRate	0.025773407542533...	0.324558842241...	1461
Row13	DistanceFro...	HourlyPayRate	0.029482168180974...	0.259763647517...	1461
Row3	Age	MonthlyInco...	0.49708287255979966	0.0	1461
Row9	BillableRate	MonthlyInco...	0.0058847109426040 7	0.822061003965...	1461
Row14	DistanceFro...	MonthlyInco...	-0.017661467007407...	0.499668192685...	1461
Row18	HourlyPayRate	MonthlyInco...	-0.018593036067930...	0.477318508284...	1461
Row4	Age	MonthlyRate	0.028942743578428...	0.268587251331...	1461
Row10	BillableRate	MonthlyRate	-0.03616138541526324	0.166844822845...	1461
Row15	DistanceFro...	MonthlyRate	0.0299245291207084 7	0.252678957239...	1461
Row19	HourlyPayRate	MonthlyRate	-0.012410578863281...	0.635280982880...	1461
Row22	MonthlyInco...	MonthlyRate	0.03504527436971207	0.180336754092...	1461
Row5	Age	PercentSala...	0.006057431049720...	0.816929589651...	1461
Row11	BillableRate	PercentSala...	0.023172702000972...	0.375780150409...	1461
Row16	DistanceFro...	PercentSala...	0.04222207668938401	0.106462493876...	1461
Row20	HourlyPayRate	PercentSala...	-0.007923082206365...	0.762044461598...	1461
Row23	MonthlyInco...	PercentSala...	-0.026353377597575...	0.313786535180...	1461
Row25	MonthlyRate	PercentSala...	-0.007355768033027...	0.778621861997...	1461

The following image is the Tableau output for correlation between continuous variables and attrition.

Continues Variables P-Value and Correlation With Attrition

Attrition	DistanceFromHome	2.064003821e-06	0.1237
	HourlyPayRate	0.920714100	0.0026
	MonthlyRate	0.948244728	0.0017
	PercentSalaryHike	0.721212613	-0.0093
	BillableRate	0.111900178	-0.0416
	MonthlyIncome	8.257611761e-08	-0.1396
	Age	1.046393067e-09	-0.1587

Correlation Value

-0.1587 0.1237

Sum of Correlation value broken down by Second column name, First column name and P Value. Colour shows sum of Correlation value. The marks are labelled by sum of Correlation value. The view is filtered on Second column name, which keeps Attrition.

The results from above image show that DistanceFromHome has a weak but significant correlation with Attrition. Also, the MonthlyIncome and Age have significant but inverse correlation with Attrition.

3.4.2 Correlation between Ordinal variables and Attrition

To carry out the correlation between ordinal variables and attrition, a total of six variables namely, Education, JobLevel, PerformanceRating, StockOptionLevel, TotalSatisfaction and Attrition were encoded. The HeatMap and composite table received from KNIME are shown below.

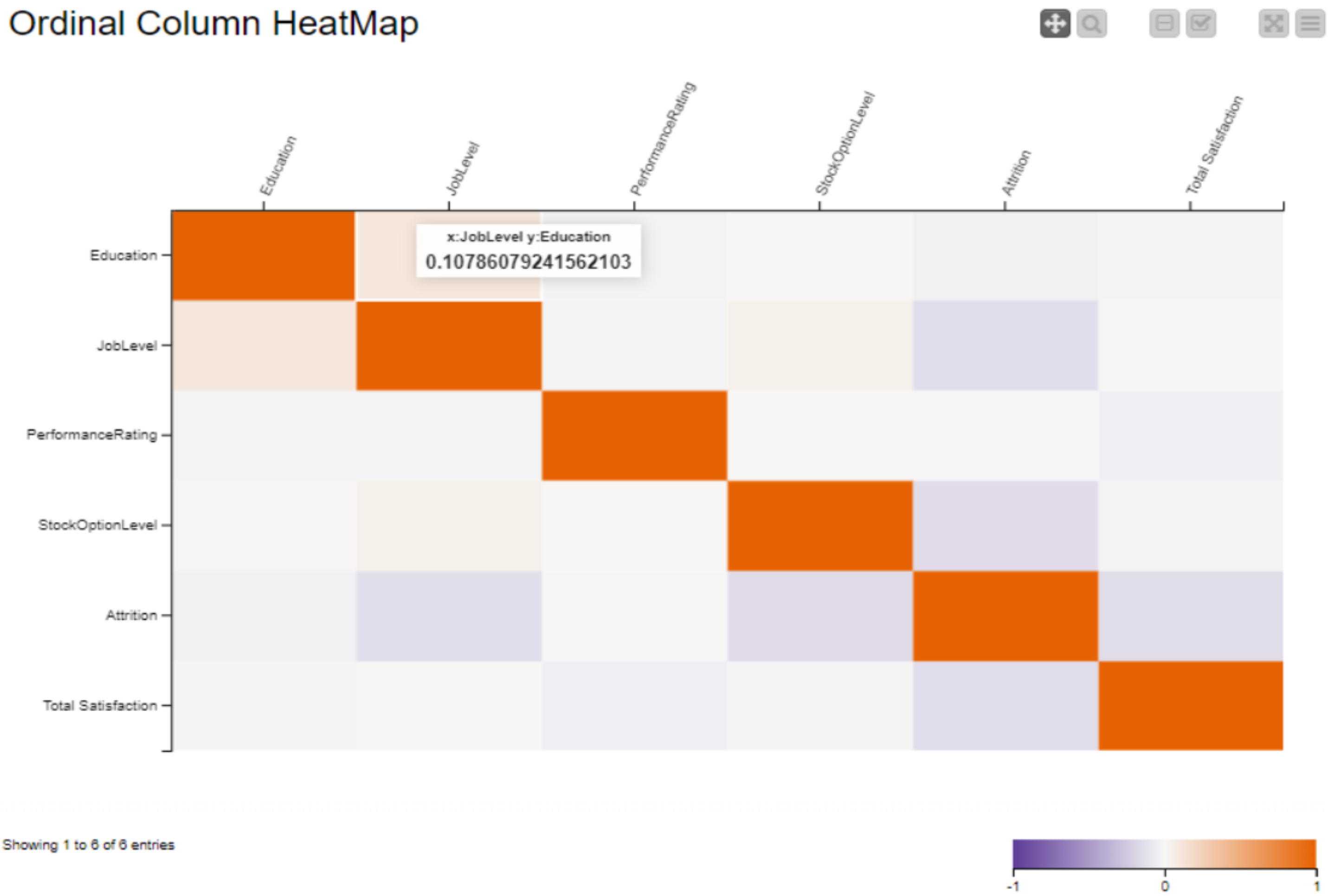
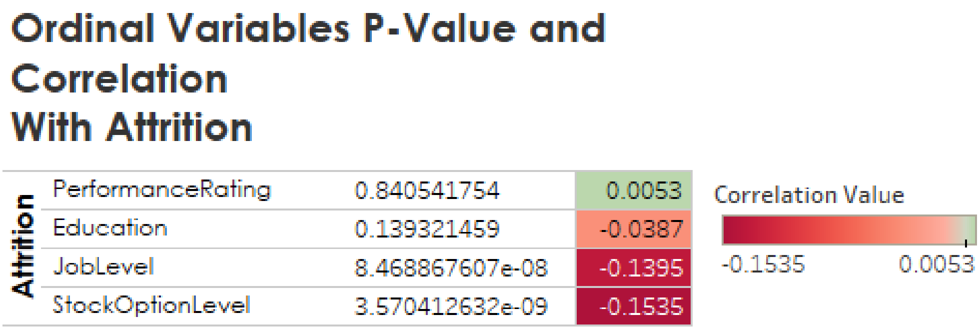


Table "default" - Rows: 15 | Spec - Columns: 5 | Properties | Flow Variables

Row ID	First col...	Sec...	Correlation value	p value	Degree...
Row3	Education	Attrition	-0.03866837841310...	0.1393214589733074	1461
Row7	JobLevel	Attrition	-0.13948523808063...	8.46886760719201E-8	1461
Row10	Performanc...	Attrition	0.00526472125329...	0.8405417537354687	1461
Row12	StockOption...	Attrition	-0.1535442402344754	3.5704126322144702...	1461
Row0	Education	JobLevel	0.10786079241562...	3.5630693605126496...	1461
Row1	Education	Performanc...	-0.02447851818186...	0.34946721156990446	1461
Row5	JobLevel	Performanc...	-0.01872311032711...	0.4742419085239189	1461
Row2	Education	StockOption...	0.01435764514419...	0.5831927591402757	1461
Row6	JobLevel	StockOption...	0.04859587573201...	0.06313185951375466	1461
Row9	Performanc...	StockOption...	0.00930099690778...	0.7222456497547405	1461
Row4	Education	Total Satisfa...	-0.01760643877812...	0.5010055409921086	1461
Row8	JobLevel	Total Satisfa...	0.00712213282928...	0.785478723149978	1461
Row11	Performanc...	Total Satisfa...	-0.04380341145712...	0.09397023993015607	1461
Row13	StockOption...	Total Satisfa...	-0.01644899999264...	0.5295680237380753	1461
Row14	Attrition	Total Satisfa...	-0.14584951975173...	2.0964603830719852...	1461

The following image is the Tableau output for correlation between ordinal variables and attrition.



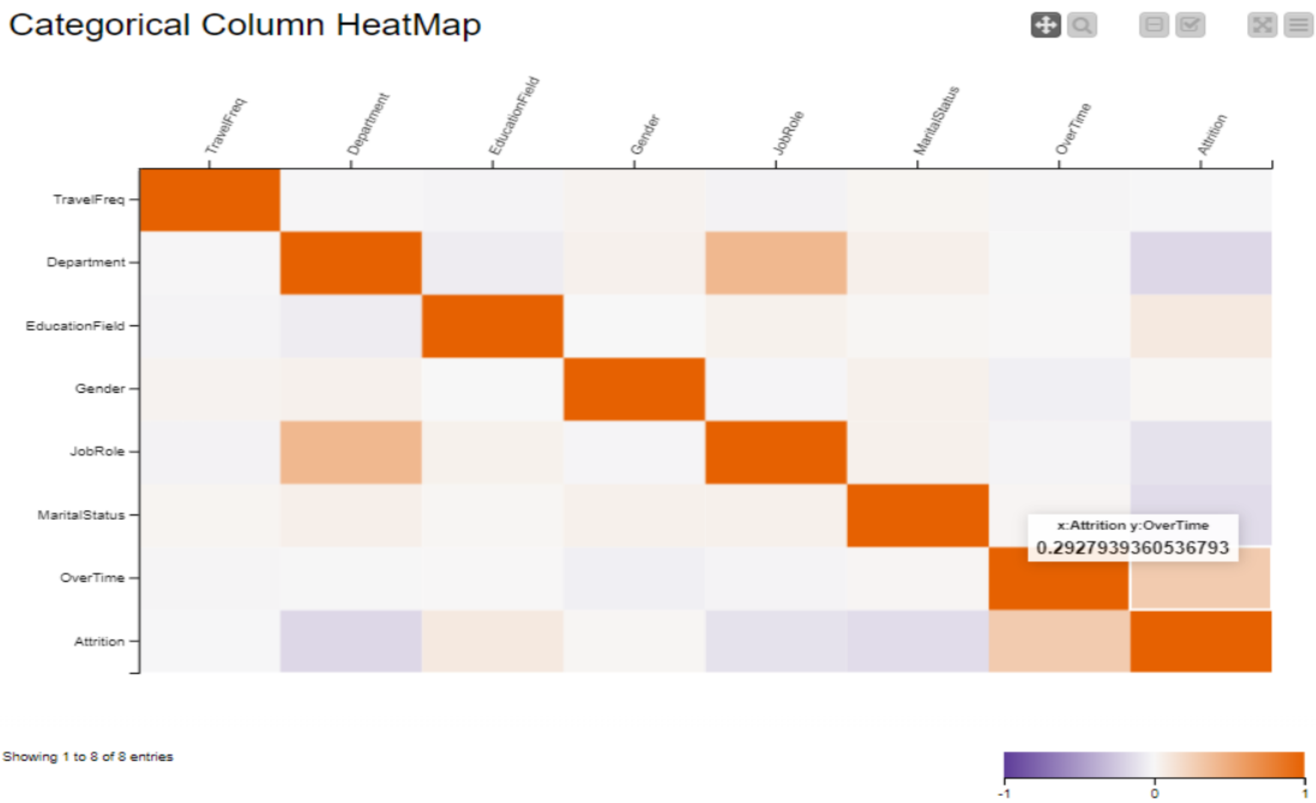
Sum of Correlation value broken down by Second column name1, First column name1 and P Value. Colour shows sum of Correlation value. The marks are labelled by sum of Correlation value. The view is filtered on Second column name1, which keeps Attrition.

Above image represents the correlation between each ordinal variable and Attrition respectively. It indicates that JobLevel and StockOptionLevel have inverse correlation with significant p-value. Wherein, the correction of PeformanceRating and Education is positive but insignificant here.

3.4.3 Correlation between Categorical variables and Attrition

Before conducting the correlation between categorical variables and attrition, a total of six variables namely, TravelFrew, EducationField, Gender, Department, JobRole, and MaritalStatus were encoded as category to number.

Also, OverTime and Attrition were encoded as 1 (Yes) and 0 (No). The HeatMap and composite table received from KNIME is shown below:



The HeatMap shows a moderate correlation between Department and JobRole, Overtime and Attrition. The output of composite table also supports these findings.

Table "default" - Rows: 28 | Spec - Columns: 5 | Properties | Flow Variables

Row ID	S First col...	S ▲ Sec...	D Correlation value	D p value	I Degree...
Row6	TravelFreq	Attrition	-0.004322841897746...	0.868782933142817	1461
Row12	Department	Attrition	-0.17145929231684287	4.0697408680734...	1461
Row17	EducationField	Attrition	0.09300809040365246	3.6780722284723...	1461
Row21	Gender	Attrition	0.01620253836606771	0.5357551109823...	1461
Row24	JobRole	Attrition	-0.11471657814056228	1.0895496136526...	1461
Row26	MaritalStatus	Attrition	-0.1422328840778823	4.6699627797073...	1461
Row27	OverTime	Attrition	0.2927939360536793	0.0	1461
Row0	TravelFreq	Department	-0.008700672147264...	0.7395000821037...	1461
Row1	TravelFreq	EducationField	-0.020017221787158...	0.4442330769976...	1461
Row7	Department	EducationField	-0.06122097761500353	0.0191885430149...	1461
Row2	TravelFreq	Gender	0.03335437069719442	0.2022944737783...	1461
Row8	Department	Gender	0.044596662276389816	0.0881615551216...	1461
Row13	EducationField	Gender	-4.800607840368111E-4	0.9853626402230...	1461
Row3	TravelFreq	JobRole	-0.02884871381755339	0.2701461379690...	1461
Row9	Department	JobRole	0.42246340615685507	0.0	1461
Row14	EducationField	JobRole	0.042748467340619024	0.1021657582389...	1461
Row18	Gender	JobRole	-0.01341952731303511	0.6080420361968...	1461
Row4	TravelFreq	MaritalStatus	0.022758232174515156	0.3843810704172...	1461
Row10	Department	MaritalStatus	0.05422777982278925	0.0380866059576...	1461
Row15	EducationField	MaritalStatus	0.01627138141893667	0.5340232445509...	1461
Row19	Gender	MaritalStatus	0.047397896366735526	0.0699244342997...	1461
Row22	JobRole	MaritalStatus	0.047342442321701	0.0702528375871...	1461
Row5	TravelFreq	OverTime	-0.01617915626989489	0.5363439704415...	1461
Row11	Department	OverTime	-0.006777743167410...	0.7956163163588...	1461
Row16	EducationField	OverTime	0.005099569294276972	0.8454799205085...	1461
Row20	Gender	OverTime	-0.04351097680719058	0.0961874580433...	1461
Row23	JobRole	OverTime	-0.019936123175496...	0.4460812698586...	1461
Row25	MaritalStatus	OverTime	0.01777287834354451	0.4969663758668...	1461

Following image represents the correlation between each categorical variable and Attrition respectively.

Categorical Variables P-Value and Correlation With Attrition

		P Value	Correlation Value
Attrition	OverTime	0	0.2928
	EducationField	0.000367807	0.0930
	Gender	0.535755111	0.0162
	TravelFreq	0.868782933	-0.0043
	JobRole	1.089549614e-05	-0.1147
	MaritalStatus	4.669962780e-08	-0.1422
	Department	4.069740868e-11	-0.1715

Correlation Value: -0.1715 to 0.2928

Sum of Correlation value broken down by Second column name1, First column name1 and P Value. Colour shows sum of Correlation value. The marks are labelled by sum of Correlation value. The view is filtered on Second column name1, which keeps Attrition.

This table finds a statistically significant moderate relationship between overtime and attrition, but JobRole, MatitalStatus, Department have negative and statically significant with Attrition. Also, the correlation between EducationField, Gender, TravelFreq and Attrition are statistically insignificant.

3.4.4 Correlation between Discrete variables and Attrition

First of all, the discrete variables were filtered out and their datatypes were corrected accordingly. Then the Attrition was encoded as 0 (No) and 1 (Yes). Here rank correlation was calculated. The HeatMap and composite table using KNIME are shown below:

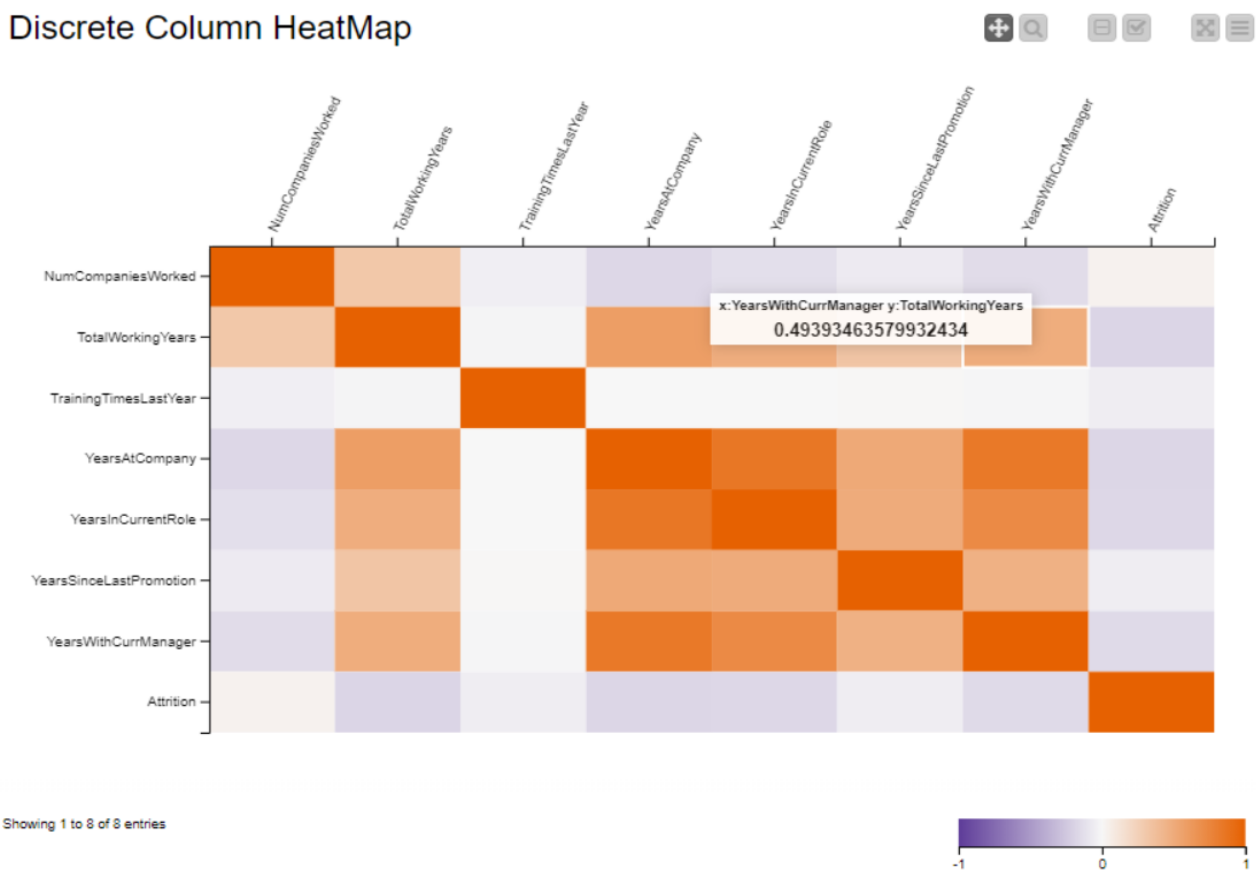


Table "default" - Rows: 28 | Spec - Columns: 5 | Properties | Flow Variables

Row ID	First column name	Second ...	Correlation value	p value	Degree...
Row6	NumCompaniesWorked	Attrition	0.03819052411129...	0.14427897...	1461
Row12	TotalWorkingYears	Attrition	-0.18376577438712...	1.40776279...	1461
Row17	TrainingTimesLastYear	Attrition	-0.05177845447522...	0.04769015...	1461
Row21	YearsAtCompany	Attrition	-0.17734784921200...	8.38351610...	1461
Row24	YearsInCurrentRole	Attrition	-0.17115171911607...	4.41322534...	1461
Row26	YearsSinceLastPromo...	Attrition	-0.05276593189706...	0.04359808...	1461
Row27	YearsWithCurrManager	Attrition	-0.15466480879886...	2.73833355...	1461
Row0	NumCompaniesWorked	TotalWorkingYears	0.31594702788210...	0.0	1461
Row1	NumCompaniesWorked	TrainingTimesLas...	-0.04676719851037...	0.07373461...	1461
Row7	TotalWorkingYears	TrainingTimesLas...	-0.01702353506197...	0.51528736...	1461
Row2	NumCompaniesWorked	YearsAtCompany	-0.1713277355785477	4.21334078...	1461
Row8	TotalWorkingYears	YearsAtCompany	0.5924980977964793	0.0	1461
Row13	TrainingTimesLastYear	YearsAtCompany	-1.10857202226474...	0.99661971...	1461
Row3	NumCompaniesWorked	YearsInCurrentR...	-0.12794057475505...	9.12489743...	1461
Row9	TotalWorkingYears	YearsInCurrentR...	0.4907429852900214	0.0	1461
Row14	TrainingTimesLastYear	YearsInCurrentR...	0.00194678095440...	0.94069274...	1461
Row18	YearsAtCompany	YearsInCurrentR...	0.8537490086128767	0.0	1461
Row4	NumCompaniesWorked	YearsSinceLastPr...	-0.06841963185748...	0.00884918...	1461
Row10	TotalWorkingYears	YearsSinceLastPr...	0.33288953741699...	0.0	1461
Row15	TrainingTimesLastYear	YearsSinceLastPr...	0.00863630050953...	0.74135821...	1461
Row19	YearsAtCompany	YearsSinceLastPr...	0.5199806333214884	0.0	1461
Row22	YearsInCurrentRole	YearsSinceLastPr...	0.5051319104084506	0.0	1461
Row5	NumCompaniesWorked	YearsWithCurrM...	-0.14450663559439...	2.82905736...	1461
Row11	TotalWorkingYears	YearsWithCurrM...	0.49393463579932...	0.0	1461
Row16	TrainingTimesLastYear	YearsWithCurrM...	-0.01325436891713...	0.61246519...	1461
Row20	YearsAtCompany	YearsWithCurrM...	0.84281366914696	0.0	1461
Row23	YearsInCurrentRole	YearsWithCurrM...	0.7244941287994073	0.0	1461
Row25	YearsSinceLastPromo...	YearsWithCurrM...	0.46810004502284...	0.0	1461

Discrete Variables P-Value and Correlation With Attrition

		P Value	Correlation Value
Attrition	NumCompaniesWorked	0.144278971	0.0382
	TrainingTimesLastYear	0.047690151	-0.0518
	YearsSinceLastPromotion	0.043598081	-0.0528
	YearsWithCurrManager	2.738333560e-09	-0.1547
	YearsInCurrentRole	4.413225341e-11	-0.1712
	YearsAtCompany	8.383516104e-12	-0.1773
	TotalWorkingYears	1.407762795e-12	-0.1838

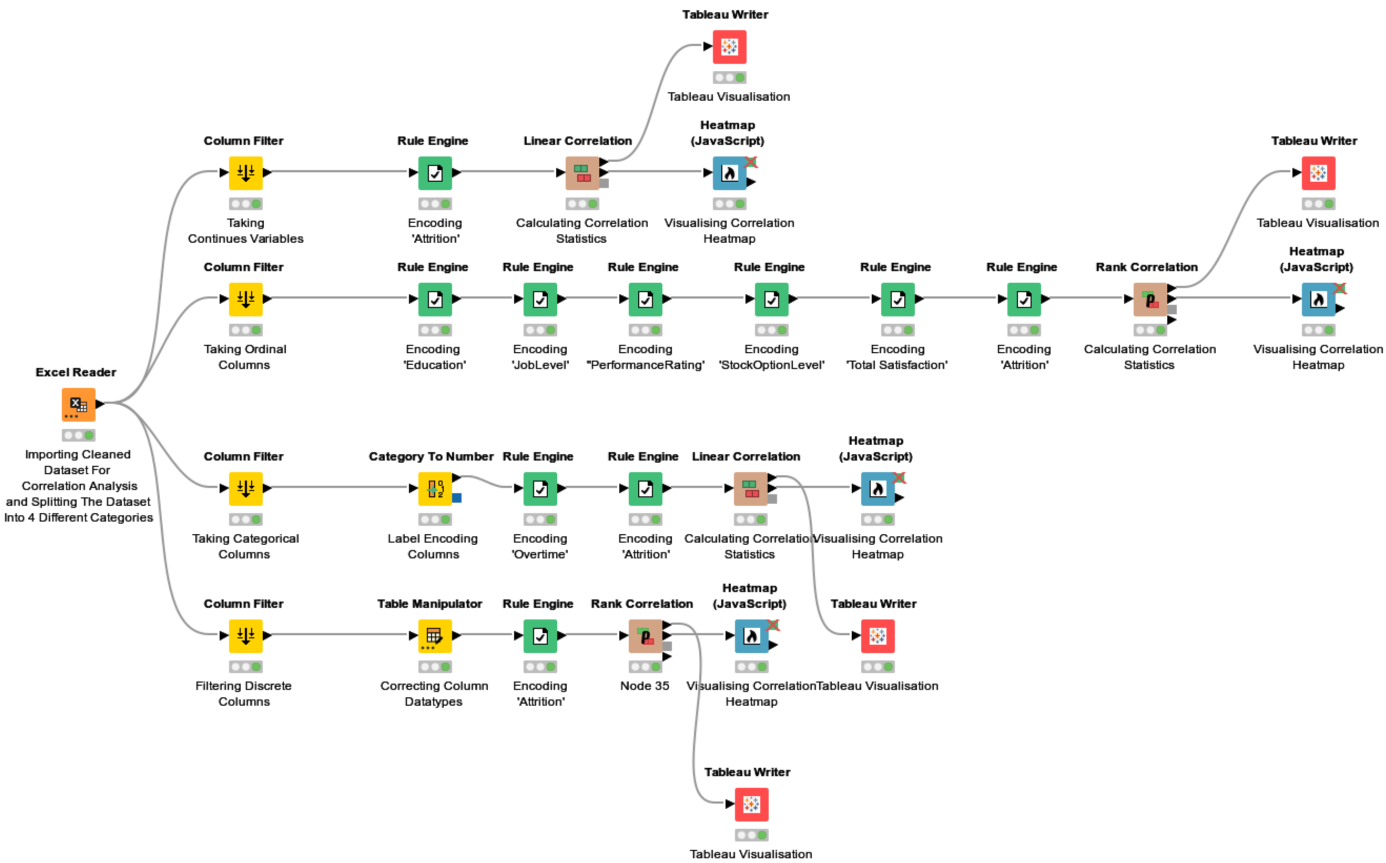
Correlation Value

-0.1838 0.0382

Sum of Correlation value broken down by Second column name1, First column name1 and P Value. Colour shows sum of Correlation value. The marks are labelled by sum of Correlation value. The view is filtered on Second column name1, which keeps Attrition.

The above table shows the Tableau output of Discrete variables and Attrition. It indicates that YearsWithCurrManager, YearsinCurrentRole, YearAtCompany, and TotalWorkingYears have inverse but statistically significant correlations with Attrition. While NumCompaniesWorked, TrainingTimeLastYear, YearsSinceLastPromotion have insignificant correlations with Attrition.

The KNIME workflow for the whole process of Attrition Correlation analysis is as follows:



Thus, the selection of factors to build the model was based on the correlation analysis and total 16 selected factors with respective correlation statistic and p-value are as follows:

Selected Features For Custom Model Building (p-Value, Corr)

Column Name	p-Value	
OverTime	0	0.2928
DistanceFromHome	2.064e-06	0.1237
EducationField	0.00036781	0.0930
TrainingTimeLastYear	0.0476	-0.0517
JobRole	1.0895e-05	-0.1147
JobLevel	8.4688e-08	-0.1394
MonthlyIncome	8.2576e-08	-0.1396
MaritalStatus	4.67e-08	-0.1422
TotalSatisfaction	1.11815e-08	-0.1486
StockOptionLevel	3.57e-09	-0.1535
YearsWithCurrManager	2.738e-09	-0.1540
Age	1.0464e-09	-0.1587
YearsInCurrentRole	4.413e-11	-0.1700
Department	4.0697e-11	-0.1715
YearsAtCompany	8.383e-12	-0.1773
TotalWorkingYears	1.407e-12	-0.1837

Correlation

-0.1837 0.2928

Sum of Correlation broken down by Column Name and sum of p-Value. Colour shows sum of Correlation. The marks are labelled by sum of Correlation.

3.4.5 Model Building and Discussion

Based on the objective of this study to build a best model to predict the employee attrition, it was estimated to prepare several models with selected factors to make a comparison between them. So, we decided to prepare four models:

a) Decision Tree Model
b) Random Forest Model
c) Logistic Regression Model
d) Custom Model

The output of each model was evaluated on following parameters:

1) Confusion Matrix

Confusion matrix is the primary means to assess errors in classification problem. It reveals the whole pattern of misclassification i.e. the actual class in which items should belong to and the wrong class to which items mistakenly belong (Beauxis-Aussalet & Hardman, 2014).

2) Accuracy

Accuracy stands for percentage of true predictions, but it is not a reliable measure (Selvaraj, 2022). So, it is essential to take other parameters into account for model evaluation.

3) Precision

Precision is the measure which shows the quality of true positive predictions with respect to total positive predictions (Davis & Goadrich, 2006).

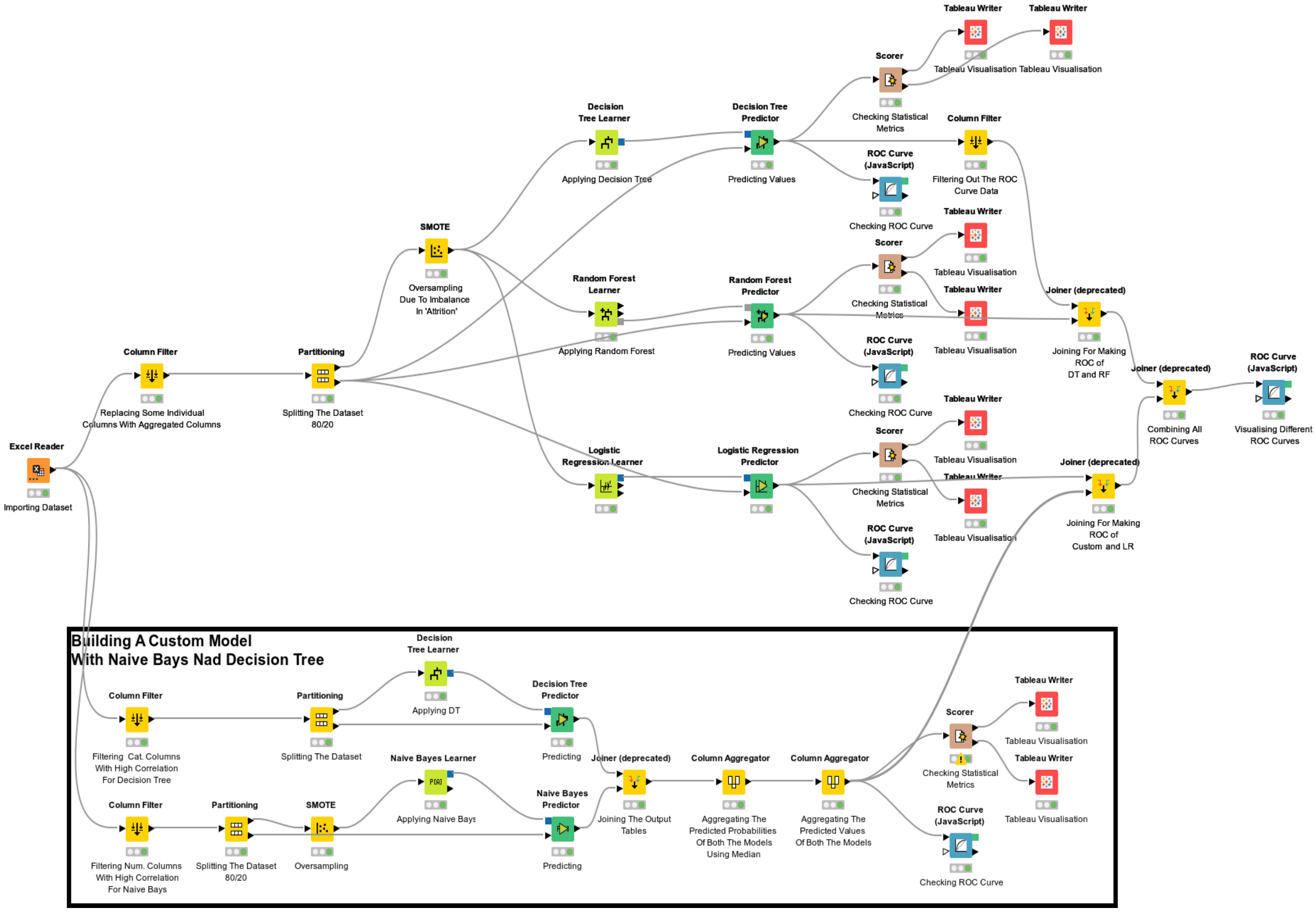
4) Cohen's kappa

It is a prominent scalar measure of accuracy proposed by Cohen (1960). Similar to correlation, kappa is a statistic to test interrater reliability (source: datatab.net)

5) ROC curve

ROC curves are the most commonly used substitute of accuracy. In ROC curve, True positives (successes) and false positives (false alarms) are plotted on two-dimensional ROC curves (Ben-David, 2008)

Following is the KNIME workflow for the model building:



a) Decision Tree Model

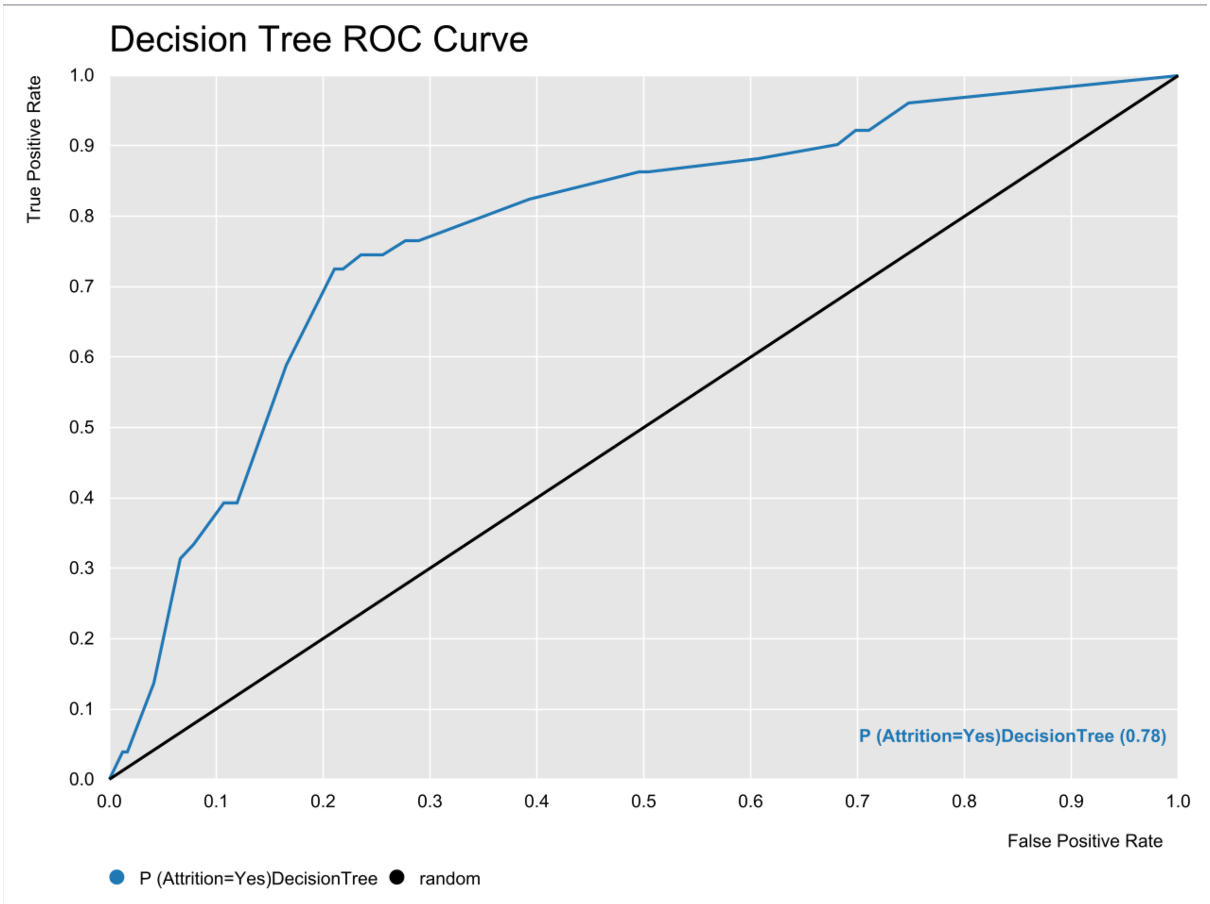
Decision tree is a popular data mining approach to create classification schemes/ prediction algorithms based on several covariates or creating prediction algorithms for a dependent variable (Song & Ying, 2015). To balance the class distribution, we also used SMOTE in decision tree (Chawla *et al*., 2002). Following is the output of the decision tree model which contains several parameters of model evaluation.

Decision Tree

		Predicted Class	
Actual Class		Yes	No
	Yes	38	13
	No	64	178

Decision Tree

TruePositives	FalsePositives	TrueNegatives	FalseNegatives	Recall	Precision	Sensitivity	Specificity	F-measure	Accuracy	Cohen's kappa
38	64	178	13	0.7451	0.372549	0.74509804	0.73553719	0.49673203		
178	13	38	64	0.73554	0.9319372	0.73553719	0.74509804	0.8221709		
									0.7372014	0.344633261



According to the above tables and ROC curve, the accuracy of the model is 73%, Cohen's Kappa is 0.344 and the p-value of ROC curve is 0.78, this indicates that 78% times the model can efficiently predict the attrition.

b) Random Forest Model

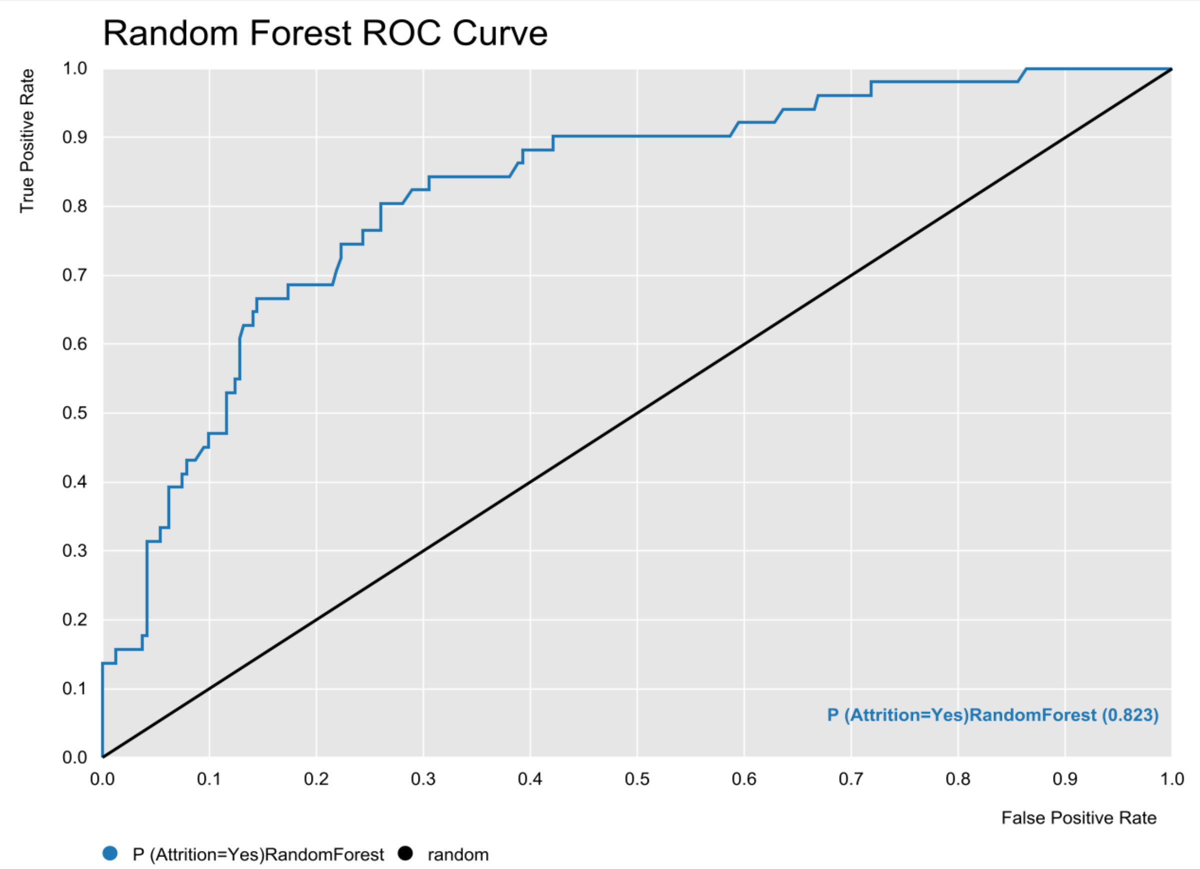
Another popular and widely applied method used in classification and prediction is known as Random Forest. It is a combined machine learning algorithm which has a series of tree classifiers. It has high classification accuracy with high tolerance for outliers and noise (Liu, Wang, & Zhang, 2012). In addition to decision tree model, we have devised random forest model, the output of the model are as following.

Random Forest

		Predicted Class	
Actual Class		Yes	No
	Yes	26	25
	No	28	214

Random Forest

TruePositives	FalsePositives	TrueNegatives	FalseNegatives	Recall	Precision	Sensitivity	Specificity	F-measure	Accuracy	Cohen's kappa
26	28	214	25	0.5098	0.4814815	0.50980392	0.88429752	0.4952381		
214	25	26	28	0.8843	0.8953975	0.88429752	0.50980392	0.88981289		
									0.8191126	0.38516055



As per the above tables and ROC curve, the accuracy of the model is 82%, Cohen's Kappa is 0.385 and the p-value of ROC curve is 0.823, this indicates that 82% times the model can efficiently predict the attrition.

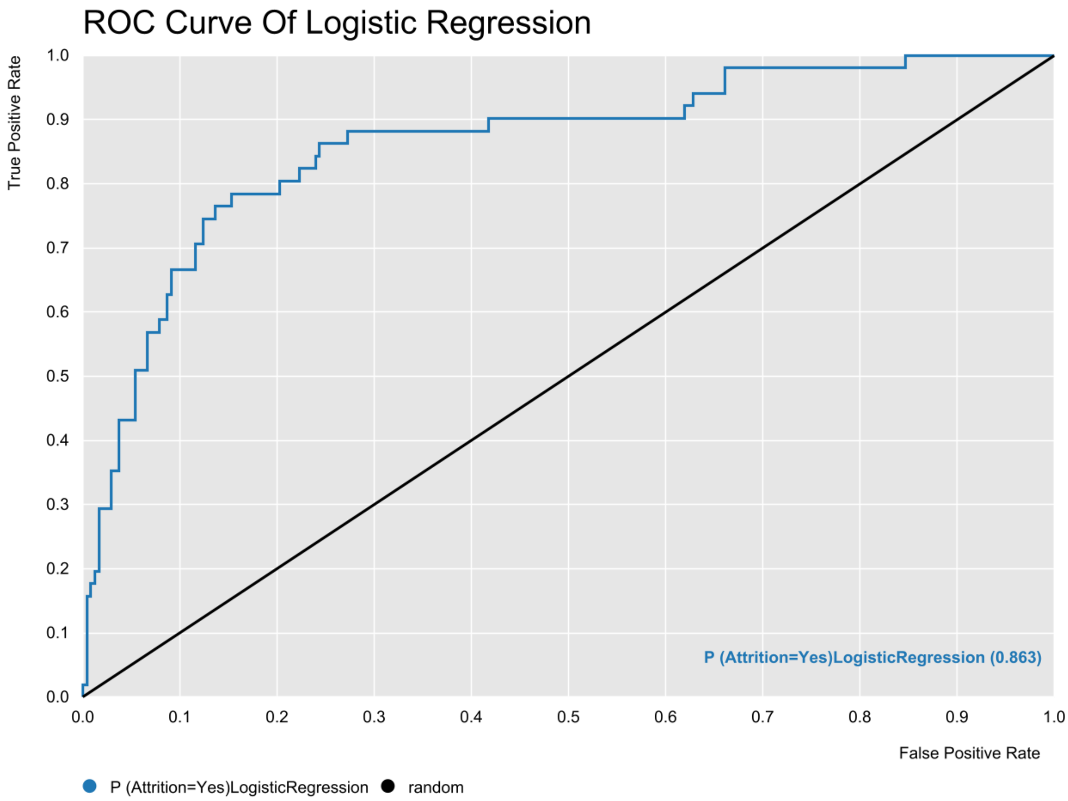
c) Logistic Regression Model

As opposed to regression model, Logistic regression is a classification problem. An extensively popular and practiced algorithm, logistic regression is simple and effective method for binary and linear classification problem (source: sciencedirect.com).

Based on the factors derived from correlation analysis, we executed logistic regression model. The output of logistic regression model is as follow.

Logistic Regression

		Predicted Class	
		Yes	No
Actual Class	Yes	40	11
	No	46	196



Logistic Regression

TruePositives	FalsePositives	TrueNegatives	FalseNegatives	Recall	Precision	Sensitivity	Specificity	F-measure	Accuracy	Cohen's kappa
40	46	196	11	0.78431	0.4651163	0.78431373	0.80991736	0.58394161		
196	11	40	46	0.80992	0.9468599	0.80991736	0.78431373	0.87305122		
									0.8054608	0.467595397

The accuracy and Cohen’s kappa of the logistic regression model is 0.805 and 0.467 which is higher than the same of previous two models. Also, the p-value of ROC curve is 0.863, this indicates that 86% times the model can efficiently predict the attrition.

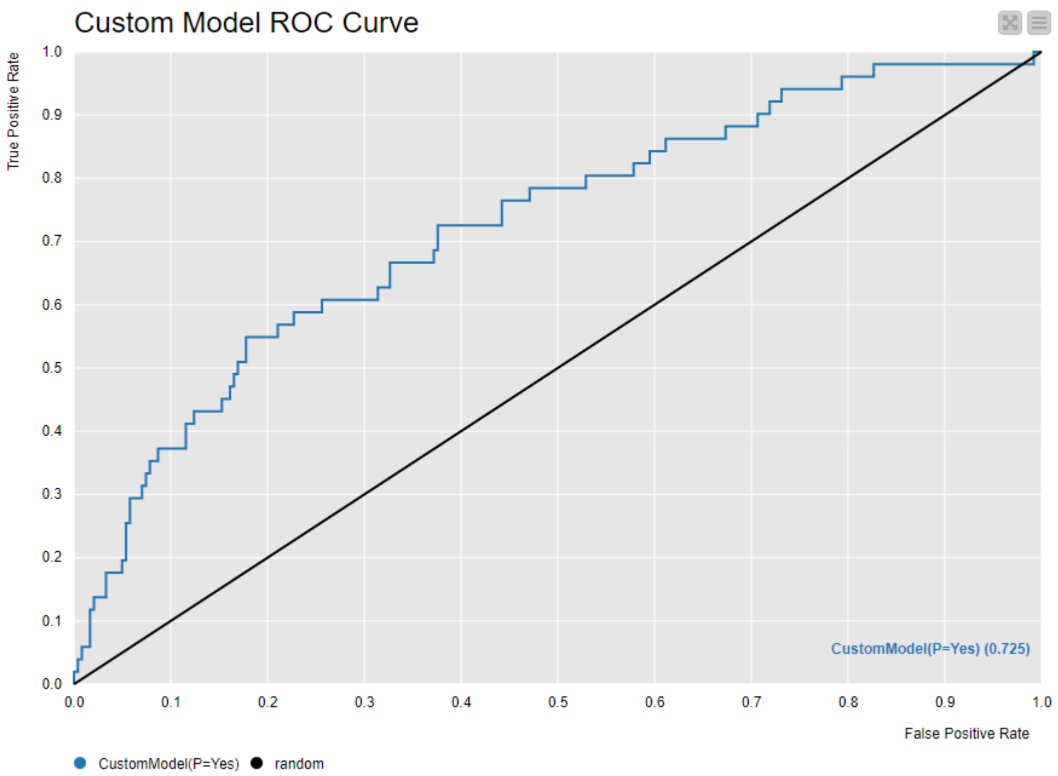
d) Custom Model

Apart from these models, we tried to develop a custom model based on Naïve bayes and Decision tree. The custom model was developed with an intention to improve the predictability power of the model. The output of the custom model is as follow:

Custom Model			
		Predicted Class	
Actual Class		Yes	No
	Yes	22	29
	No	30	212

Custom Model

TruePositives	FalsePositives	TrueNegatives	FalseNegatives	Recall	Precision	Sensitivity	Specificity	F-measure	Accuracy	Cohen's kappa
22	30	212	29	0.43137	0.4230769	0.43137255	0.87603306	0.42718447		
212	29	22	30	0.87603	0.879668	0.87603306	0.43137255	0.87784679		
									0.7986348	0.305045226

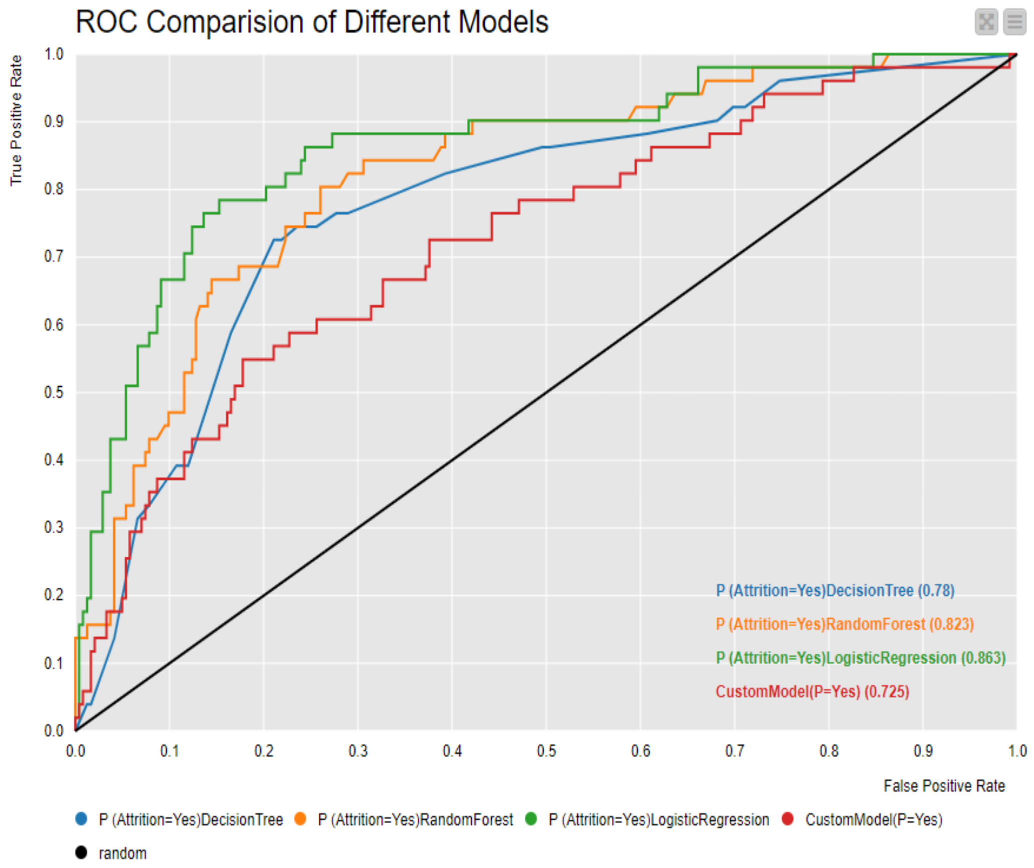


The accuracy and Cohen's kappa of the custom model are 0.798 and 0.305 respectively. Similar to the previous models, the p-value of ROC curve is 0.725, this indicates that 72.5% times the model can efficiently predict the attrition.

To conclude about the best model, we compared the miscellaneous parameters of four models. The various parameters and ROC curves of the four models are shown below.

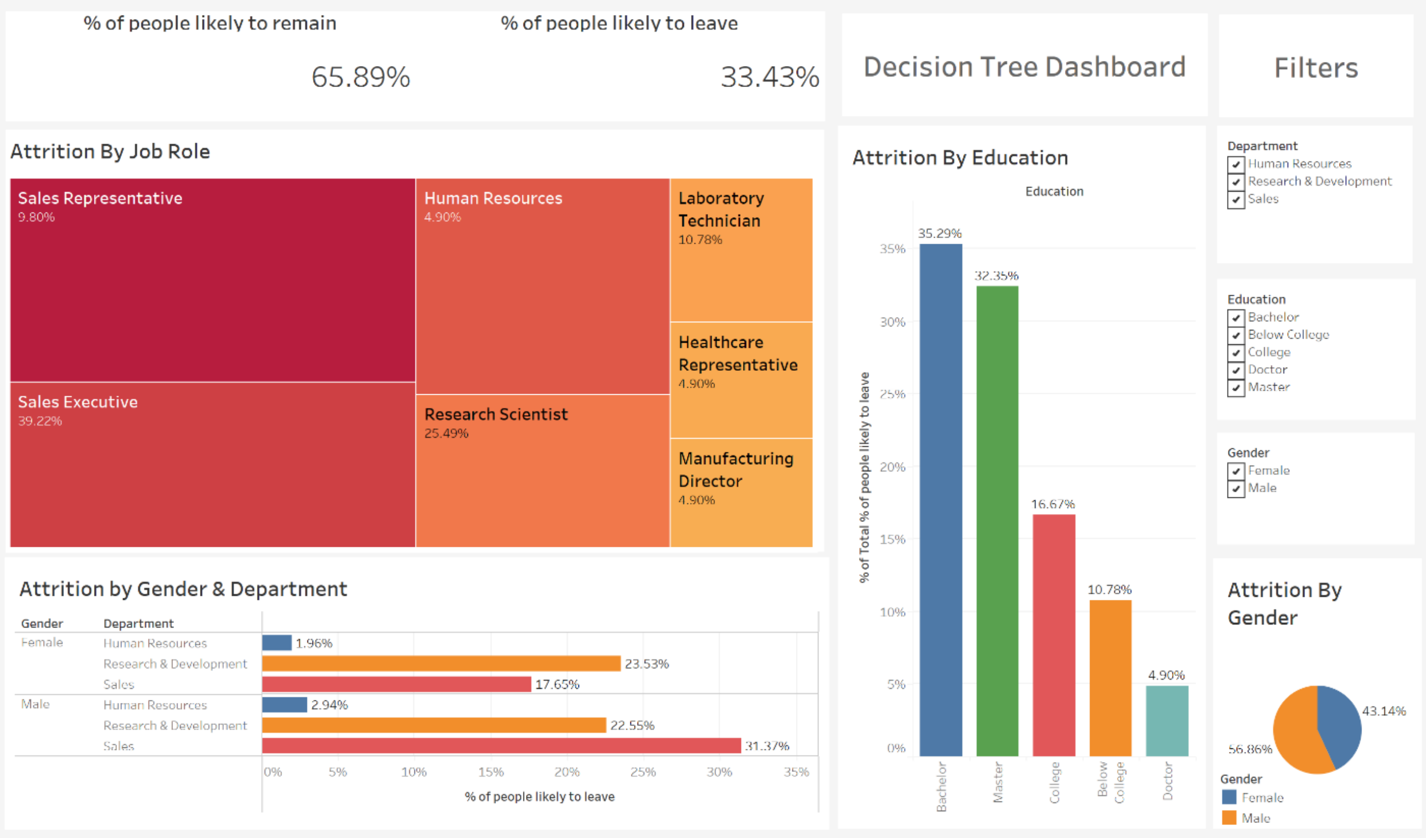
Model Comparision

Model	Accuracy	Recall	Precision	Sensitivity	Specificity	F-measure	Cohen's kappa
Logistic Regression	0.805	0.784	0.465	0.784	0.810	0.584	0.468
Decision Tree	0.737	0.745	0.373	0.745	0.736	0.497	0.345
Random Forest	0.819	0.510	0.481	0.510	0.884	0.495	0.385
Custom Model	0.798	0.431	0.423	0.431	0.876	0.427	0.305



From the observation of comparison table, it is explicit the measures of Accuracy, Recall, Sensitivity, and Cohen's kappa are highest for Logistic regression model, which was followed by Random Forest Model. The findings are supported by the comparison of ROC curves of different models. It shows that ROC curve for logistic regression has higher predictability, after that Random Forest model has better predictability. Thus, we can conclude that Logistic regression and Random Forest model are the two best models to predict employee attrition. Also, the Custom Model has better predictability over Decision Tree Model.

TABLEAU DASHBOARD OF THE MODEL BUILT FOR BETTER VISUALISATION



This dashboard provides comprehensive overview of attrition factors. Key findings here are:

- From the built model we can observe that the overall likelihood of employees remaining with the company is 65.89%, while 33.43% are likely to leave.
- Sales Executives and Research Scientists have the highest attrition rates.
- Higher attrition is seen in employees with Bachelor and College education levels.
- Male employees in the Sales department show a significantly higher attrition rate compared to other departments and genders.

It can be seen that, the company should focus retention efforts on Sales roles, address the factors contributing to higher turnover among male employees and consider strategies for engaging employees with Bachelor and Master education levels to reduce attrition.

4. CONCLUSION

The key objective of this project was to leverage predictive analytics to model employee attrition for an organization. The presence of high attrition levels impacts productivity, profitability as well as continuity for business operations. Being able to predict the risk of attrition can enable targeted human resource interventions towards employee retention.

A comprehensive dataset with 35 variables on 1467 employees was utilised in the project. Using KNIME and Tableau, exploratory analysis is conducted to understand trends and correlations between different factors and attrition.

The key conclusions from the analysis are:

- The logistic regression model performed the best with accuracy of 80.5% and Kappa of 0.467 on test data
- Random forest was comparable with 82% accuracy but lower Kappa at 0.385
- Custom model combining Naïve Bayes and Decision Tree achieved accuracy of 79.8%
- Traditional tree models did well but had limitations in predictive performance
- Attrition was highest amongst young employees of ages 26-34 years and for sales representatives
- Factors like monthly income, total years with company, job level etc. had significant negative correlation with attrition

The findings align with previous research which has found salary, job position and age to be key predictors in churn models (Li, 2022). The use of ensemble methods like random forest rather than single decision trees has also been established to enhance accuracy of results (Verbeke et al., 2012).
The use of advanced analytics techniques enabled robust insights into the factors influencing attrition. The logistic regression and random forest models can serve as useful tools to identify employees most at risk of quitting.

There is scope to enhance model accuracy further through techniques like neural networks, gradient boosting etc. Overall, the project successfully demonstrated the applicability of machine learning techniques for employee turnover predictions. The logistic model can serve as a robust diagnostic tool for identifying vulnerable employees, so that proactive retention strategies can be executed by HR departments. With further refinement and updates, predictive capabilities can become integral for data-driven human capital decisions in modern day organizations.

5. RECOMMENDATIONS

1. **Implement the logistic regression model for predictions**: The logistic model had superior performance over other techniques with accuracy of 80.5%. By operationalizing this model, each current employee can be scored monthly on their risk of attrition. High risk employees can be flagged for preventive actions.

2. **Focus retention efforts on key segments:** Sales personnel and millennials saw significantly higher turnover. Customized interventions like flexible work, career growth opportunities, incentive programs should be explored for these vulnerable segments (Yang et al., 2020).

3. **Strengthen compensation and promotion practices:** Higher pay and faster promotions were linked to lower quit rates. Benchmarking with industry standards and establishing clear advancement pathways can enhance retention (Regus, 2021).

4. **Conduct exit interviews & analysis:** Departing employees can provide useful insights into pain points regarding company culture, policies, work environment. Text/sentiment analytics on exit interviews can supplement predictive models.

5. **Continually track and optimize attrition KPIs:** With real-time tracking of monthly/quarterly attrition rates across different employee groups, trends can be monitored and addressed through timely interventions. Models also need periodic re-validation on new data.

6. **Focus on employee engagement initiatives:** Enhanced empowerment, manager rapport, growth opportunities and work-life balance have proven links with reducing turnover. Investing in these areas can complement analytics efforts (Wang et al., 2020).

Integrating the attrition prediction model with strategic HR practices targeting high attrition employee clusters can formulate an effective retention strategy. The recommendations encompass both HR analytics for predictive capabilities as well as qualitative initiatives towards building an engaging workplace. By combining databased diagnostics with managerial wisdom, significant reductions in avoidable employee turnover can be achieved.

The use of analytics, data-driven insights coupled with qualitative measures can help formulate an effective retention strategy.

There is scope for further work by incorporating text analytics on exit interviews and employee feedback. With continuous model validation and enhancement, predictive capabilities can become more robust.

6. LIMITATIONS

1. **Limited data variables:** While the dataset had 35 variables, there may be other relevant variables that impact attrition which were not captured. As noted by Dutta (2014), lack of relevant variables like employee engagement levels, organizational culture, manager relationships etc. in HR analytics studies can restrict model performance due to omitted variable bias issues. Incorporating such attributes could improve model accuracy.

2. **Sample size**: The analysis was done on data for 1,467 employees. While this is reasonably good, a larger dataset with higher number of churned employees could have aided more robust validation of the machine learning models. Smaller sample size has been attributed to overfitting problems in churn prediction models as discussed by Verbeke et al. (2012). Larger volumes help stabilize findings.

3. **Model overfitting**: As seen in the decision tree results, overfitting the training data was a possibility leading to lower generalizability. More sophisticated validation through k-fold cross validation could help negate this (Verbeke et al. 2012).

4. **Lack of qualitative data:** Hard numbers can only reveal a partial picture. Inclusion of text or survey data capturing employee attitudes, sentiment would enable a more holistic diagnosis of turnover drivers. This Qualitative feedback from employees enables a better understanding of employee attrition drivers highlighting an area of improvement in analytics (Ramkissoon, 2013).

While the analysis provides meaningful indicative findings, incorporating the above enhancements can enrich the quality of results even further. As part of future work, overcoming some of these limitations through expanded data collection, additional variables and sophisticated analytics would be valuable.

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8. APPENDIX

MINUTES OF THE MEETINGS

A) Date: 28th December 2023 time: 11 am.

Attendees: Mohd. Kaif, Mohd. Saad, Prafful Ramesh, Shubhanshu and Tabish Raza.

1) Discussed general guidelines and requirements of the assignment.

2) Thorough inspection of the datatypes and methodology was discussed.

3) Clear division of roles and responsibility of each member was decided.

B) Date: 3rd January 2024 time: 11 am.

Attendees: Mohd. Kaif, Mohd. Saad, Prafful Ramesh, Shubhanshu and Tabish Raza.

1) Discussed the ongoing progress of the assignment.

2) Every member has presented his task.

3) Some doubts were discussed and clarified by other members.

4) Discussion over Assignment report writing guidelines.

5) Further roles and responsibilities for report writing was allocated to each member.

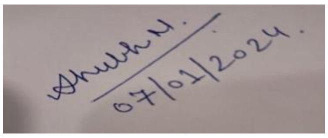
DECLARATION OF EQUAL CONTRIBUTION

We, the undersigned members of the group, hereby declare that we have all contributed equally to the *completion* of this assignment. We collectively affirm that each member has actively and significantly participated in the conceptualization, research, analysis and writing processes of this project. We understand and agree that the mark awarded for this assignment will be received equally by all members, reflecting our joint effort and collaboration.

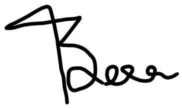
Mohammad Kaif Tahir

Prafful Ramesh

Saad Haroon



Shubhanshu Naik



Tabish Raza

Date: [08/01/2024]